

606030



Test Booklet No.

ENGLISH

Test Booklet Code

10

NEET RANKERS TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. 50 questions in each subject are divided into **two Sections (A and B)** as per details given below:
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries 4 marks. For each correct response, the candidate will get **4 marks**. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
- Use Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The **CODE** for this Booklet is **10**. **Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): _____

Roll Number : In figures _____

: In words _____

Centre of Examination (in Capitals): _____

Candidate's Signature: _____ Invigilator's Signature: _____

Facsimile Signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

1. Velocity of a body moving along a straight line with uniform acceleration (a) reduces by $\frac{3}{4}$ of its initial velocity in time t_0 .
The total time of motion of the body till its velocity becomes zero is
(1) $\frac{4}{3}t_0$ (2) $\frac{3}{2}t_0$
(3) $\frac{5}{3}t_0$ (4) $\frac{8}{3}t_0$
2. A vessel contains 1 mole of O_2 gas (molar mass 32) at a temperature T . The pressure of the gas is p . An identical vessel containing one mole of He gas (molar mass 4) at temperature $2T$ has a pressure of
(1) $p/8$ (2) p
(3) $2p$ (4) $8p$
3. The conduction current in ideal case through a circuit is zero when charge on capacitor is
(1) zero
(2) maximum
(3) any transient value
(4) depends on capacitor used
4. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: If amplitude of SHM is increased period of SHM will increase.
Reason R: If amplitude is increased, body have to travel more distance in one complete oscillation.
In the light of the above statements, choose the correct answer from the options given below.
(1) Both **Assertion (A)** and **Reason (R)** are true and **Reason (R)** is the correct explanation of **Assertion (A)**.
(2) Both **Assertion (A)** and **Reason (R)** are true but **Reason (R)** is not the correct explanation of **Assertion (A)**.
(3) **Assertion (A)** is true and **Reason (R)** is false.
(4) **Assertion (A)** is false and **Reason (R)** is true.
5. A flywheel of moment of inertia 2 kg-m^2 is rotated at a speed of 30 rad/s . A tangential force at the rim stops the wheel in 15 second. Average torque of the force is
(1) 4 N-m (2) 2 N-m
(3) 8 N-m (4) 1 N-m
6. A circular current carrying coil has a radius R . The distance from the centre of the coil on the axis of the coil, where the magnetic induction is $(1/8)^{\text{th}}$ of its value at the centre of coil is
(1) $\sqrt{3}R$ (2) $R/\sqrt{3}$
(3) $(2\sqrt{3})R$ (4) $R/2\sqrt{3}$
7. In an $L-C-R$ series AC circuit, at resonance
(1) the capacitive reactance is more than the inductive reactance
(2) the capacitive reactance equals the inductive reactance
(3) the capacitive reactance is less than the inductive reactance
(4) the current is minimum
8. The logic gates having output 1 for the inputs of 1 and 0 are
(1) AND and OR (2) OR and NOR
(3) NAND and NOR (4) NAND and OR
9. What is the value of $[(5.0 \times 10^{-6}) \times (5.0 \times 10^{-8})]$ with due regards to significant digits?
(1) 25×10^{-14} (2) 25.0×10^{-14}
(3) 2.05×10^{-13} (4) 250×10^{-15}
10. The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied?
(1) Length = 50 cm, diameter = 0.5 mm
(2) Length = 100 cm, diameter = 1 mm
(3) Length = 200 cm, diameter = 2 mm
(4) Length = 300 cm, diameter = 3 mm

11. A body of mass 10 kg is placed on rough surface, pulled by a force F making an angle of 30° above the horizontal. If the angle of friction is also 30° , then the minimum magnitude of force F required to move the body is equal to (take, $g = 10\text{ms}^{-2}$)

(1) 100 N (2) $50\sqrt{2}$ N
(3) $100\sqrt{2}$ N (4) 50 N

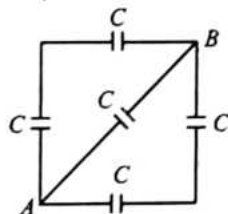
12. Two solenoids acting as short bar magnets P and Q are arranged such that their centres are on the X -axis and are separated by a large distance. The magnetic axes of P and Q are along X and Y -axes, respectively. At a point R , mid-way between their centres, if B is the magnitude of induction due to Q , then the magnitude of total induction at R due to the both magnets is

(1) $3B$ (2) $\sqrt{5}B$
(3) $\frac{\sqrt{5}}{2}B$ (4) B

13. An electron is accelerated from rest through a potential difference of V volt. If the de-Broglie wavelength of the electron is 1.227×10^{-2} nm, the potential difference is

(1) 10^2 V (2) 10^3 V
(3) 10^4 V (4) 10 V

14. In the figure shown, the effective capacitance between the points A and B , if each has capacitance C , is



(1) $2C$ (2) $C/5$
(3) $5C$ (4) $C/2$

15. In double slit experiment, for light of which colour, the fringe width will be minimum?

(1) Violet (2) Red
(3) Green (4) Yellow

16. In an electromagnetic wave, the electric and magnetic fields are 100Vm^{-1} and 0.265 T, respectively. The maximum rate of energy flow is

(1) $21 \times 10^6 \times \text{Wm}^{-2}$
(2) $36.5 \times 10^6 \times \text{Wm}^{-2}$
(3) $46.7 \times 10^6 \times \text{Wm}^{-2}$
(4) $42 \times 10^6 \times \text{Wm}^{-2}$

17. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: If radius of earth suddenly shrinks to half its present without changing its mass value, then the period of an earth's satellite will not change.

Reason R: Time period of a satellite does not depend upon the mass of earth.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both **Assertion (A)** and **Reason (R)** are true and **Reason (R)** is the correct explanation of **Assertion (A)**.
(2) Both **Assertion (A)** and **Reason (R)** are true but **Reason (R)** is not the correct explanation of **Assertion (A)**.
(3) **Assertion (A)** is true and **Reason (R)** is false.
(4) **Assertion (A)** is false and **Reason (R)** is true.

18. Given below are two statements.

Statement I: Dimensionally incorrect equation can never be physically correct.

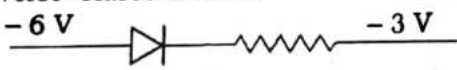
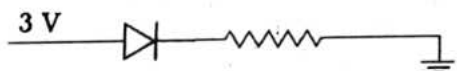
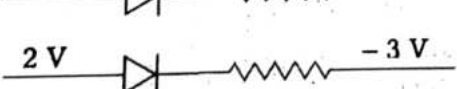
Statement II: Physically correct equations must be dimensionally correct.

In the light of the above statements, choose the most appropriate answer from the options given below.

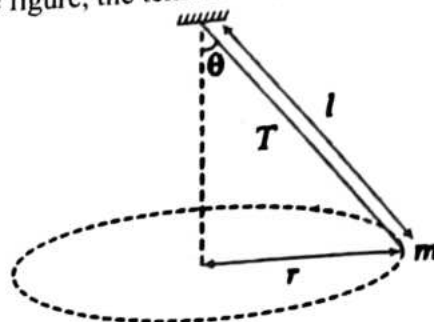
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

19. A car of mass 1000 kg negotiates a banked curve of radius 90 m on a frictionless road. If the banking angle is 45° , the speed of the car is
 (1) 20 ms^{-1} (2) 30 ms^{-1}
 (3) 5 ms^{-1} (4) 10 ms^{-1}
20. A galvanometer of 25Ω and having full scale deflection for a current of 10 mA is changed into voltmeter of range 100 V by connecting a resistance R in series with the galvanometer. The resistance R (in ohm) is
 (1) 10000 (2) 975
 (3) 10025 (4) 9975
21. For the Bohr's first orbit of circumference $2\pi r$, the de-Broglie wavelength of revolving electron will be
 (1) $2\pi r$ (2) πr
 (3) $\frac{1}{2\pi r}$ (4) $\frac{1}{4\pi r}$
22. A convex lens of refractive index $3/2$ has a power of 2.5 D. If it is placed in a liquid of refractive index 2, the new power of the lens is
 (1) 2.5 D (2) -2.5 D
 (3) 1.25 D (4) -1.25 D
23. If an alternating voltage is represented as $E = 141 \sin(628t)$, then the rms value of the voltage and the frequency are respectively
 (1) 141V, 628 Hz
 (2) 100 V, 50 Hz
 (3) 100 V, 100 Hz
 (4) 141 V, 100 Hz

24. A reverse-biased diode is

- (1)  (2) 
 (3)  (4) 

25. A string of length l fixed at one end carries a mass m at the other end. The string makes $\frac{2}{\pi} \text{ rev s}^{-1}$ around the axis through the fixed end as shown in the figure, the tension in the string is

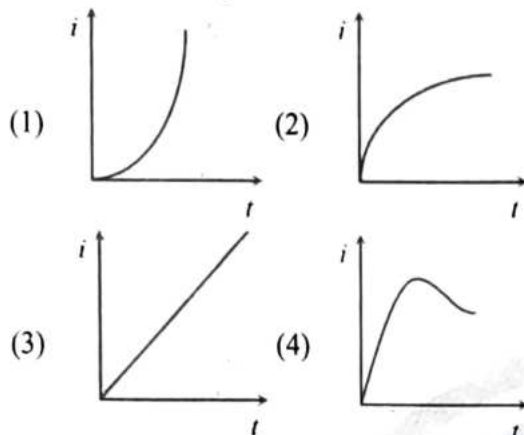


- (1) $16 ml$ (2) $4 ml$
 (3) $8 ml$ (4) $2 ml$
26. An electric dipole of moment $\mathbf{p}$ is placed normal to the lines of force of electric intensity $\mathbf{E}$, then the work done in deflecting it through an angle of 180° is
 (1) pE (2) $+2pE$
 (3) $-2pE$ (4) zero
27. A sphere is released on a smooth inclined plane from the top. When it moves down, its angular momentum is:
 A. Conserved about every point
 B. Conserved about the point of contact only
 C. Conserved about any point on a line parallel to the inclined plane and passing through the centre of the ball.
- Choose the most appropriate answer from the options given below:
 (1) A only (2) C only
 (3) B and C only (4) A, B and C

28. Electric field at a far away distance r on the axis of a dipole is E_0 . What is the magnitude of electric field at a distance $2r$ on perpendicular bisector?

- (1) $\frac{E_0}{16}$ (2) $\frac{E_0}{24}$
 (3) $\frac{E_0}{8}$ (4) $\frac{E_0}{18}$

29. When a battery is connected across a series combination of self-inductance L and resistance R , then the variation in the current i with time t is best represented by

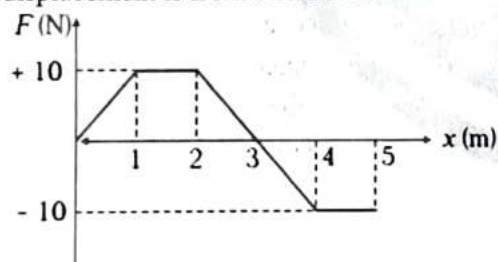


30. A 50 kg boy stands on a platform spring scale in a lift that is going down with a constant speed 3 ms^{-1} . If the lift is brought to rest by a constant deceleration in a distance of 9m, what does the scale read during this period?

(Take, $g = 9.8 \text{ ms}^{-2}$)

- (1) 500 N (2) 465 N
(3) 515 N (4) zero

31. A position dependent force F is acting on a particle and its force-position curve is shown in the figure. Work done on the particle, when its displacement is from 0 to 5 m is



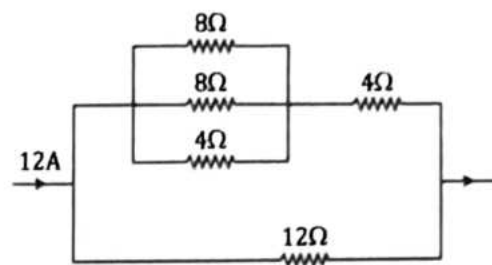
- (1) 35 J (2) 25 J
(3) 15 J (4) 5 J

32. A slit of width d is illuminated by white light. For red light ($\lambda = 6500 \text{ \AA}$), the first minima is obtained at $\theta = 30^\circ$.

Then, the value of d will be

- (1) $3250 \text{ \AA}$ (2) $13 \times 10^{-4} \text{ mm}$
(3) 1.24 microns (4) $26 \times 10^{-4} \text{ cm}$

33. In the given circuit, find voltage across 12Ω resistance.



- (1) 12 V (2) 36 V
(3) 72 V (4) 48 V

34. An experiment measures quantities a , b and c , and X is calculated from $X = ab^2/c^3$. If the percentage error in a , b and c are $\pm 1\%$, $\pm 3\%$ and $\pm 2\%$ respectively, the maximum percentage error in X will be:

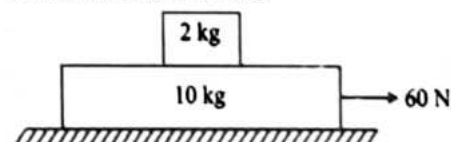
- (1) $\pm 13\%$ (2) $\pm 7\%$
(3) $\pm 4\%$ (4) $\pm 1\%$

35. When two displacements represented by $y_1 = a \sin(\omega t)$ and $y_2 = b \cos(\omega t)$ are superimposed, the motion is

- (1) not a simple harmonic
(2) simple harmonic with amplitude $\frac{a}{b}$
(3) simple harmonic with amplitude $\sqrt{a^2 + b^2}$
(4) simple harmonic with amplitude $\frac{(a+b)}{2}$

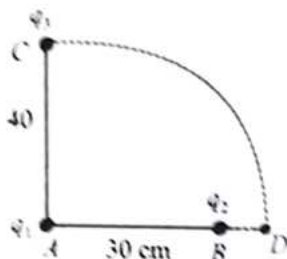
Physics : Section-B (Q. No. 36 to 50)

36. A plank of mass 10 kg and a block of mass 2 kg are placed on a horizontal plane as shown in the figure. There is no friction between plane and plank. The coefficient of friction between block and plank is 0.5. A force of 60 N is applied on plank horizontally. In first 2 s the work done by friction on the block is

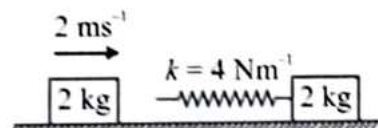


- (1) -100 J (2) 100 J
(3) zero (4) 200 J

37. Two charges q_1 and q_2 are placed 30 cm apart as shown in the figure. A third charge q_1 is moved along the arc of a circle of radius 40 cm from C to D. The change in the potential energy of the system is $\frac{q_1}{4\pi\epsilon_0} k$, where k is



- (1) $8q_2$ (2) $8q_1$
(3) $6q_2$ (4) $6q_1$
38. If I_1 is the moment of inertia of a thin rod about an axis perpendicular to its length and passing through its centre of mass and I_2 is the moment of inertia of the ring about an axis perpendicular to plane of ring and passing through its centre formed by bending the rod, then
- (1) $\frac{I_1}{I_2} = \frac{3}{\pi^2}$ (2) $\frac{I_1}{I_2} = \frac{2}{\pi^2}$
(3) $\frac{I_1}{I_2} = \frac{\pi^2}{2}$ (4) $\frac{I_1}{I_2} = \frac{\pi^2}{3}$
39. In the arrangement shown in figure match the following.

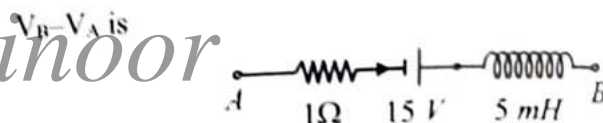


- | List-I | List-II |
|---|----------------|
| (A) Velocities of centre of mass | (I) 2 SI unit |
| (B) momentum of combined mass when compression in the spring is maximum | (II) 1 SI unit |

- (C) Half of maximum compression in the spring
 $\left(\frac{X_{\max}}{2}\right)$
- (D) Maximum potential energy stored in the spring
- Choose the **correct** answer from the options given below.

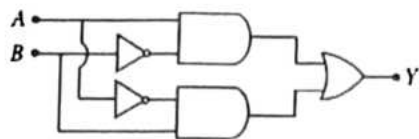
- (1) A-I, B-II, C-III, D-IV
(2) A-III, B-I, C-II, D-IV
(3) A-IV, B-II, C-I, D-III
(4) A-II, B-III, C-IV, D-I

40. The network shown in the figure is a part of a complete circuit. If at a certain instant the current I is 5 A and decreasing at the rate of 10^3 As^{-1} , then



- (1) 5 V (2) 10 V
(3) 15 V (4) 20 V
41. A microscope is focused on an ink mark on the top of a table. If we place a glass slab of 3 cm thick on it, how should the microscope be moved to focus the ink spot again? The refractive index of glass is 1.5.
- (1) 2 cm upwards (2) 2 cm downwards
(3) 1 cm upwards (4) 1 cm downwards
42. The escape velocity from earth is v_e . A body is projected with velocity $2v_e$. With what constant velocity will it move in the inter planetary space?
- (1) v_e (2) $\sqrt{2}v_e$
(3) $\sqrt{3}v_e$ (4) $\sqrt{5}v_e$

43. In the combination of the following gates, the output Y can be written in terms of inputs A and B as



- (1) $\overline{A \cdot B} + A \cdot B$ (2) $A \cdot \overline{B} + \overline{A} \cdot B$
 (3) $\overline{A \cdot B}$ (4) $\overline{A + B}$

44. When the light of frequency $2\nu_0$ (where, ν_0 is threshold frequency), is incident on a metal plate, the maximum velocity of electrons emitted is ν_1 . When the frequency of the incident radiation is increased to $5\nu_0$, the maximum velocity of electrons emitted from the same plate is ν_2 . The ratio of $\nu_1 : \nu_2$ is

- (1) 4 : 1 (2) 1 : 4
 (3) 1 : 2 (4) 2 : 1

45. Force constants of two wires A and B of the same material are k and $2k$ respectively. If the two wires are stretched equally, then the ratio of work

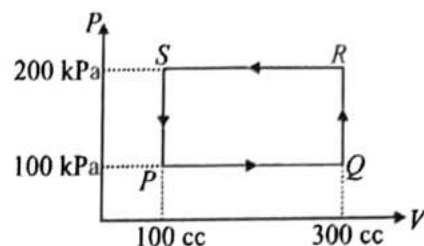
done in stretching $\left(\frac{W_A}{W_B}\right)$ is

- (1) $\frac{1}{3}$ (2) $\frac{1}{2}$
 (3) $\frac{3}{2}$ (4) $\frac{1}{4}$

46. Current i is carried in a wire of length L . If the wire is turned into a circular coil, the maximum magnitude of torque in a given magnetic field B will be

- (1) $\frac{L^2 B^2}{2}$ (2) $\frac{L^2 B}{2}$
 (3) $\frac{L^2 i B}{4\pi}$ (4) $\frac{L^2 B}{4\pi}$

47. A thermodynamic system is taken through the cyclic process $PQRSP$. The net work done by the system is



- (1) 20 J (2) -20 J
 (3) 400 J (4) -400 J

48. A body of mass m is attached to the lower end of a spring whose upper end is fixed. The spring has negligible mass. When the mass m is slightly pulled down and released, it oscillates with a time period of 3 s. When the mass m is increased by 1 kg, the time period of oscillations becomes 5 s. The value of m in kg is

- (1) $\frac{3}{4}$ (2) $\frac{4}{3}$
 (3) $\frac{16}{9}$ (4) $\frac{9}{16}$

49. A bar magnet of magnetic moment M is axially cut into two equal parts. If these two pieces are arranged perpendicular to each other, the resultant magnetic moment is M^2 . Then, the value of $\frac{M_1}{M_2}$ is

- (1) $\frac{1}{2\sqrt{2}}$ (2) 1
 (3) $\frac{1}{\sqrt{2}}$ (4) $\sqrt{2}$

50. Two point charges A and B , having charges $+Q$ and $-Q$ respectively, are placed at certain distance apart and force acting between them is F . If 25% charge of A is transferred to B , then force between the charges becomes

- (1) $\frac{9F}{16}$ (2) $\frac{16F}{9}$
 (3) $\frac{4F}{3}$ (4) F

Chemistry : Section-A (Q. No. 51 to 85)

51. Temperature dependence of reaction rates can be studied by plotting a graph between:
- Concentration of reactants and temperature
 - Concentration of products and temperature
 - Rate constant and temperature
 - Rate of catalysis and temperature
52. The oxidation state of Fe in FeO is:
- +2
 - $+\frac{200}{93}$
 - $-\frac{93}{200}$
 - +3
53. Vapours of an alcohol X when passed over hot reduced copper, produce an alkene, the alcohol is:
- Primary alcohol
 - Secondary alcohol
 - Tertiary alcohol
 - Dihydric alcohol
54. A compound with molecular formula C_3H_6O , not gives silver mirror with Tollens' reagent but forms oxime with hydroxyl amine. Compound will be;
- $CH_2=CH-CH_2-OH$
 - CH_2CH_2CHO
 - $CH_2=CH-CH_3$
 - CH_3COCH_3
55. Correct statement(s) are
- H_3PO_4 is a tribasic acid.
 - H_3PO_4 contains three P – OH groups.
 - Hypophosphorous acid is a good oxidising agent.
 - Hypophosphorous acid contains two P – H bonds.
- Only A and B
 - A, B and D
 - A and C
 - B and C

56. Match List-I with List-II.

List-I (Compound)	List-II (lone pair on central atom)
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- | | |
|--------------|------------|
| (A) XeF_2 | (I) Zero |
| (B) SO_2 | (II) One |
| (C) H_2O | (III) Two |
| (D) $BeCl_2$ | (IV) Three |

Choose the **correct** answer from the options given below.

- A-IV, B-III, C-II, D-I
- A-IV, B-II, C-III, D-I
- A-I, B-II, C-III, D-IV
- A-III, B-IV, C-I, D-II

57. Mn^{2+} compounds are more stable than Fe^{2+} compound towards oxidation to their +3 state, because:

- Mn^{2+} is more stable with high 3rd ionisation energy
- Mn^{2+} is bigger in size
- Mn^{2+} has completely filled *d*-orbitals
- Mn^{2+} does not exist.

58. Match List-I with List-II.

List-I (Process)	List-II (Conditions)
---------------------	-------------------------

- | | |
|------------------------|---|
| (A) Reversible process | (I) process carried out Very slow |
| (B) Isothermal | (II) process carried out at constant pressure |
| (C) Isobaric | (III) process carried out at constant temperature |
| (D) Adiabatic | (IV) Heat change during process is zero |

Choose the **correct** answer from the options given below.

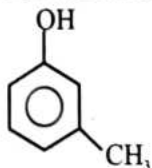
- A-I, B-II, C-III, D-IV
- A-II, B-III, C-I, D-IV
- A-III, B-IV, C-I, D-II
- A-I, B-III, C-II, D-IV

59. For a reaction: $A+B \rightarrow \text{Product}$, the rate of reaction at Various concentration are given below:

Exp No.	[A]	[B]	Rate (mol dm^{-3})
I.	0.2 M	0.2 M	2
II.	0.2 M	0.4 M	4
III.	0.6 M	0.4 M	36

The rate law expression for the above data is;

- (1) $r = k[A]^2[B]$
 (2) $r = k[A][B]^2$
 (3) $r = k[A]^3[B]$
 (4) $r = k[A]^2[B]^2$
60. IUPAC name of following compound is;

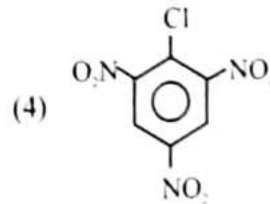
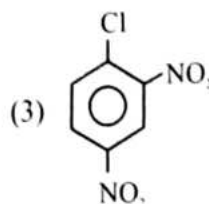
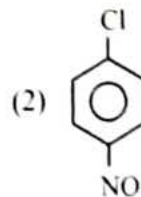
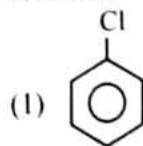


- (1) 3-Hydroxytoluene
 (2) 3-Methylphenol
 (3) 1-Hydroxy-3-methylbenzene
 (4) m-Cresol
61. Ions which have same number of unpaired electron;
- A. Ti^{3+} B. V^{2+}
 C. Ni^{2+} D. Cu^{2+}

Choose the *most appropriate* answer from the options given below.

- (1) A, B and C only
 (2) A and B only
 (3) A and C only
 (4) A and D only
62. Which of the following elements has the most negative electron gain enthalpy?
- (1) $[\text{Ne}]3s^23p^3$
 (2) $[\text{Ne}]3s^23p^4$
 (3) $[\text{Ne}]3s^23p^5$
 (4) $[\text{Ne}]3s^23p^6$

63. Which of the following haloarenes is least reactive toward nucleophilic substitution reaction?



64. Given below are two statements.

Statement I: Central atom of coordination compound also referred as Lewis acid.

Statement II: Ligands are also referred as Lewis base.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.
65. Which among the following molecule is paramagnetic in nature?
- (1) H_2 (2) N_2
 (3) O_2 (4) C_2
66. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R).
- Assertion (A):** The first ionisation enthalpy of Gallium is higher than Aluminium.
Reason (R): Due to poor shielding effect of d-orbital in Gallium, the ionisation enthalpy increases.



In the light of the above statements, choose the **correct** answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

67. Match List-I with List-II.

List-I (Reaction)	List-II (Element or Compound)
(A) $\text{CH}_2 = \text{CH}_2 \rightarrow \text{CH}_3 - \text{CH}_3$	(I) NaOH
(B) $\text{C}_6\text{H}_5\text{OH} \rightarrow \text{C}_6\text{H}_5\text{ONa}$	(II) H_2/Ni
(C) $\text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{COONa}$	(III) LiAlH_4
(D) $\text{CH}_3\text{COOH} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$	(IV) NaHCO_3

Choose the **correct** answer from the options given below.

- (1) A-II, B-I, C-IV, D-III
- (2) A-I, B-II, C-III, D-IV
- (3) A-III, B-I, C-IV, D-II
- (4) A-IV, B-III, C-I, D-II

68. The product of the following reaction is ;



- (1) $\begin{array}{c} \text{Br} \\ | \\ \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{COOH} \end{array}$
- (2) $\begin{array}{c} \text{Br} \\ | \\ \text{CH}_3 - \text{CH} - \text{CH}_2 - \text{COOH} \end{array}$
- (3) $\begin{array}{c} \text{Br} \\ | \\ \text{CH}_3 - \text{CH}_2 - \text{CH} - \text{COOH} \end{array}$
- (4) $\begin{array}{c} \text{O} \\ || \\ \text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{C} - \text{Br} \end{array}$

69. Tollen's reagent and Fehling's solution are weak oxidising agent.

- A. Tollen's reagent is ammoniacal silver nitrate solution
- B. Fehling solution is an aqueous copper sulphate solution

C. Fehling solution is an alkaline sodium potassium tartrate

Choose the *most appropriate* answer from the options given below.

- (1) A, B and C
- (2) A only
- (3) A and B only
- (4) A and C only

70. Match List-I with List-II.

List-I (Compounds)	List-II (Total no. of bond between carbon atoms)
(A) Butane	(I) one σ -bond and one π -bond
(B) Ethyne	(II) Two σ -bonds
(C) Propane	(III) Three σ -bond
(D) Ethene	(IV) one σ -bond and two π -bond

Choose the **correct** answer from the options given below.

- (1) A-III, B-II, C-I, D-IV
- (2) A-III, B-IV, C-II, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-I, B-II, C-III, D-IV

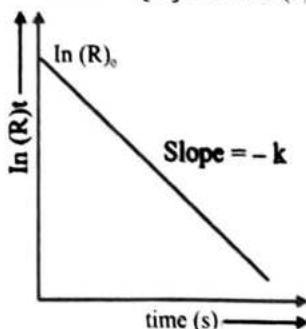
71. Which of the following orders is **correct** order for the bond dissociation enthalpy of halogen molecules?

- (1) $\text{Br}_2 > \text{I}_2 > \text{F}_2 > \text{Cl}_2$
- (2) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
- (3) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$
- (4) $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$

72. The vapour density of undecomposed N_2O_4 is 46. When heated, vapour density decreases to 24.5 due to its dissociation to NO_2 . The percentage dissociation of N_2O_4 at the final temperature is:

- (1) 87.7%
- (2) 60.0%
- (3) 40.3%
- (4) 70.6%

73. For a chemical reaction, variation in the concentration $[R]$ vs time (s) plot is given:



- (1) Zero order
(2) First order
(3) Second order
(4) Third order
74. 4.5 moles each of hydrogen and iodine are heated in a sealed 10 litre vessel. At equilibrium, 3 moles of HI are formed. The equilibrium constant for;
 $H_2(g) + I_2(g) \rightarrow 2HI(g)$ is:
 (1) 1
 (2) 10
 (3) 5
 (4) 0.33
75. Given below are two statements.
Statement I: First four members of alkanes are gases, and five to Seventeen are liquid.
Statement II: With increase in the number of carbon atoms in chain vander waal force increases.
 In the light of the above statements, choose the most appropriate answer from the options given below.
 (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

76. In S_N1 the first step involves the formation of:
 (1) Free radical
 (2) Carbanion
 (3) Carbocation
 (4) None of these
77. If the total energy of an electron is -0.54 eV in hydrogen atom then find out P.E. of the electron in that orbit:
 (1) -1.08 eV
 (2) 0.27 eV
 (3) -0.54 eV
 (4) 2.16 eV
78. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Maltose is reducing sugar but sucrose is non reducing Sugar.
Reason (R): Reducing group of glucose and fructose are involved in glycosidic linkage in sucrose.
 In the light of the above statements, choose the correct answer from the options given below.
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
79. Which method is used to purify liquids having very high boiling points and those, which decomposes at below their boiling points is:
 (1) Fractional distillation
 (2) Distillation under high pressure
 (3) Distillation under reduced pressure
 (4) Simple distillation

80. Match List-I with List-II.

List-I (Complexes)	List-II (Hybridisation)
(A) $[\text{Ni}(\text{CN})_4]^{2-}$	(I) sp^3
(B) $[\text{CuCl}_4]^{2-}$	(II) dsp^2
(C) $[\text{Co}(\text{NH}_3)_6]^{3+}$	(III) sp^3d^2
(D) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(IV) d^2sp^3

Choose the **correct** answer from the options given below.

- (1) A-II, B-I, C-III, D-IV
- (2) A-I, B-II, C-III, D-IV
- (3) A-II, B-I, C-IV, D-III
- (4) A-I, B-II, C-IV, D-III

81. Given below are two statements.

Statement I: 2° amine is more basic than 3° amine in aqueous state.

Statement II: 3° amine is more basic than 2° amine in aqueous state.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

82. Which of the following element has second ionisation enthalpy less than first ionisation enthalpy?

- (1) Mg
- (2) Ne
- (3) C
- (4) None

83. Number of carbon atoms in 0.6 mol of $\text{C}_6\text{H}_{12}\text{O}_6$ is:

- (1) $0.6 N_A$
- (2) $1.2 N_A$
- (3) $2.4 N_A$
- (4) $3.6 N_A$

84. When chloroform and water mixed together to prepare solution then;

- (1) temperature of solution increases
- (2) temperature of solution decreases
- (3) volume of solution increases
- (4) solution show positive deviation from Raoult's law

85. Which type of hydroxide generally formed by Lanthanoids ?

- (1) $\text{Ln}(\text{OH})_3$
- (2) $\text{Ln}(\text{OH})_2$
- (3) $\text{Ln}(\text{OH})_4$
- (4) $\text{Ln}(\text{OH})_6$

Chemistry : Section-B (Q. No. 86 to 100)

86. A compound is made up of two elements X and Y where X (atomic mass 10) is 50% and Y (atomic mass 20) is 50%. What will be its empirical formula of compound?

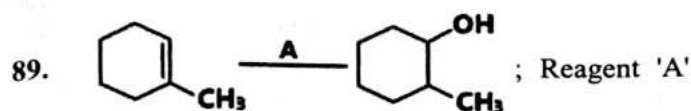
- (1) XY_2
- (2) X_2Y
- (3) XY_3
- (4) X_2Y_3

87. The **correct** increasing order of boiling points of alcohols is;

- (1) propan-1-ol < butan-2-ol < butan-1-ol < pentan-1-ol
- (2) propan-1-ol < butan-1-ol < butan-2-ol < pentan-1-ol
- (3) pentan-1-ol < butan-2-ol < butan-1-ol < propan-1-ol
- (4) pentan-1-ol < butan-1-ol < butan-2-ol < propan-1-ol

88. What would be the osmotic pressure of 0.1 M K_2SO_4 solution (90% dissociated) at 27°C :
(Given $R = 0.0821 \text{ L atm K}^{-1}\text{mol}^{-1}$)

- (1) 6.89 atm
- (2) 0.689 atm
- (3) 0.344 atm
- (4) 3.4 atm



- (1) $\text{BH}_3/\text{THF} + \text{H}_2\text{O}_2/\text{OH}^-$
- (2) $\text{H}_2\text{O}/\text{H}^+$
- (3) $\text{Hg}(\text{OCOCH}_3)_2/\text{NaBH}_4$
- (4) $\text{Cl}_2/\text{aq. NaOH}$

90. Given below are two statements.

Statement I: When AgNO_3 is added to a solution of $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, 2 moles of AgCl precipitate is obtained.

Statement II: Only secondary valencies are ionisable.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are incorrect.
- (4) Statement I is correct but Statement II is incorrect.

91. How many moles of $\text{K}_2\text{Cr}_2\text{O}_7$ can be reduced by 1 mole of Sn^{2+}

- (1) $\frac{1}{3}$
- (2) $\frac{1}{6}$
- (3) $\frac{2}{3}$
- (4) 1

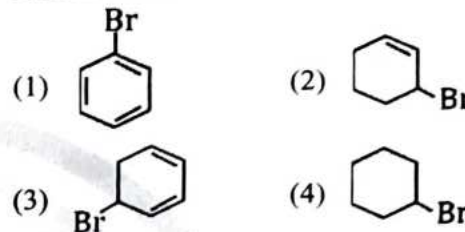
92. NH_3 has abnormally high boiling point because it has :

- (1) Alkaline nature
- (2) Distorted shape
- (3) sp^3 - Hybridisation
- (4) Hydrogen bonding

93. Group reagent for the precipitation of group II basic radicals for the qualitative analysis is :

- (1) dil. $\text{HCl} + \text{H}_2\text{S}$
- (2) $(\text{NH}_4)_2\text{CO}_3$ solution
- (3) dil. HCl
- (4) $\text{NH}_4\text{Cl}(\text{solid}) + \text{NH}_4\text{OH}(\text{solution}) + \text{H}_2\text{S}$

94. Which of the following dehydrohalogenated to a maximum extent?



95. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): On electrolysis of aq. NaCl solution, H_2 liberated at Cathode.

Reason (R): Reduction potential of H^+ ion is lower than Na^+ ion.

In the light of the above statements, choose the correct answer from the options given below.

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

96. The dissociation of CO_2 can be expressed as $2\text{CO}_2(\text{g}) \rightleftharpoons 2\text{CO}(\text{g}) + \text{O}_2(\text{g})$.

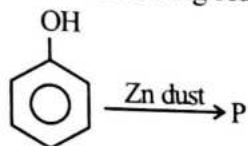
If 2 moles of CO_2 is taken initially and 40% of the CO_2 is dissociated completely, what is the total number of moles at equilibrium?

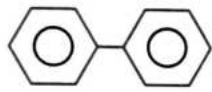
- (1) 2.4
- (2) 2.2
- (3) 1.2
- (4) 5

97. 3L gas expanded against constant external pressure of 1 atm to 15 L, the work done during the process is; (Given that 1 Latm = 101.3 J)
- $-1.215 \times 10^3 \text{ J}$
 - $+12.15 \times 10^3 \text{ J}$
 - $+121.5 \times 10^3 \text{ J}$
 - $+1.215 \times 10^3 \text{ J}$

98. On passing a current through KCl solution, 19.5 g of potassium is deposited. If the same quantity of electricity is passed through a solution of aluminium chloride, the amount of aluminium deposited is;
- 4.5 g
 - 9.0 g
 - 13.5 g
 - 27 g

99. In the following reaction product P is;



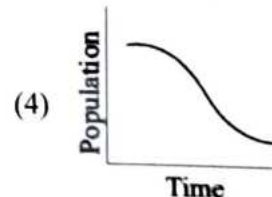
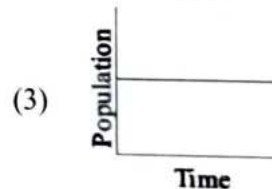
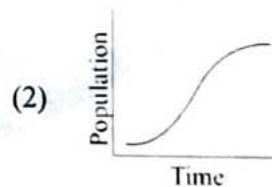
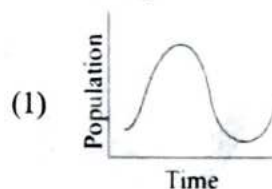
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100. Number of oxygen atoms in 8 g of ozone is:

- 6.02×10^{23}
- $\frac{6.02 \times 10^{23}}{2}$
- $\frac{6.02 \times 10^{23}}{3}$
- $\frac{6.02 \times 10^{23}}{6}$

Botany : Section-A (Q. No. 101 to 135)

101. A population which has equal number of individuals in pre-reproductive age group and reproductive age group means that:
- it is a declining population.
 - it is a stable population.
 - the population is expanding.
 - population will first expand and then decline.
102. Which of the following steps is **not** involved in DNA fingerprinting?
- Digestion of DNA by restriction endonucleases.
 - Gel electrophoresis.
 - Blotting to synthetic membranes.
 - Reverse transcription of hybridized DNA.
103. In a given population of 2000 individuals, 160 birth and 250 deaths were reported over a given period of time. Which of the following graphs will correspond to it?



104. Which of the following statements are correct?

- (a) Mendel conducted artificial pollination or cross pollination experiments using several true-breeding pure lines of pea plants.
- (b) Mendel conducted hybridisation experiments on garden peas for 7 years.
- (c) Mendel conducted hybridisation experiments on garden pea (*Pisum sativum*).
- (d) Mendel selected 7 true-breeding pea plant varieties.

- (1) (a), (b) and (c)
- (2) (a) and (d)
- (3) (a), (c) and (d)
- (4) (b), (c) and (d)

105. Read the following four statements.

- (a) Dough, which is used for making foods such as dosa and idli, is fermented by fungi and algae.
- (b) Toddy, a traditional drink of southern India is made by fermenting sap from palms.
- (c) Large holes in Swiss cheese are due to production of large amount of methane by *Propionibacterium sharmanii*.
- (d) In our stomach, lactic acid bacteria play very beneficial role in checking disease-causing microbes.

Choose the incorrect statements.

- (1) (a) & (c) (2) (a) & (b)
- (3) (b) & (c) (4) (c) & (d)

106. Which of the following are features unique to kingdom Fungi?

- (a) Nuclear envelope
- (b) Cellulosic cell walls
- (c) Chitinous cell walls
- (d) Membrane-bound organelles

Options:

- (1) (c) only
- (2) (a), (b) and (d) only
- (3) (c) and (d) only
- (4) (a), (c) and (d) only

107. Sequence the following steps involved in the decomposition process.

- (a) Catabolism (b) Humification
- (c) Fragmentation (d) Mineralisation
- (e) Leaching

Options:

- (1) (b)-(a)-(c)-(e)-(d)
- (2) (c)-(b)-(e)-(d)-(a)
- (3) (e)-(c)-(d)-(a)-(b)
- (4) (c)-(e)-(a)-(b)-(d)

108. Match List-I with List-II.

List-I (Placentation)	List-II (Example)
(A) Axile	(I) Mustard
(B) Parietal	(II) Lemon
(C) Basal	(III) Primrose
(D) Free central	(IV) Sunflower

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-III, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-II, B-I, C-IV, D-III
- (4) A-I, B-IV, C-III, D-II

109. All the following statements hold **true** for dicotyledonous stem, **except**:

- (1) the cells of the endodermis are rich in starch grains.
- (2) water-containing cavities are present within the vascular bundles.
- (3) medullary rays are present in between the vascular bundles.
- (4) 'ring' arrangement of vascular bundles is seen.

110. A single turn of citric acid cycle yields:

- (1) 2 FADH₂, 2 NADH + H⁺, 2 GTP.
- (2) 1 FADH₂, 2 NADH + H⁺, 1 GTP.
- (3) 1 FADH₂, 3 NADH + H⁺, 1 GTP.
- (4) 2 ATP, 3 NADH + H⁺, 1 GTP.

111. Identify the **incorrect** statement from the following.
- (1) ATP is not formed during photorespiration.
 - (2) C_4 -pathway of photosynthetic CO_2 fixation is also known as Hatch and Slack pathway.
 - (3) NADPH is formed during photorespiration.
 - (4) In C_4 plants, phosphoenol pyruvate (PEP) acts as primary CO_2 acceptor.
112. The **correct** sequence of structural genes in *lac* operon is:
- (1) gene *y*, gene *z*, gene *a*.
 - (2) gene *z*, gene *y*, gene *a*.
 - (3) gene *y*, gene *a*, gene *z*.
 - (4) gene *a*, gene *z*, gene *y*.
113. Given below are two statements:
- Statement I:** The ovary in hypogynous flowers is said to be superior.
- Statement II:** The ovary in perigynous flowers is said to be half inferior.
- In the light of the above statements, choose the *most appropriate* answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.
114. Choose the **incorrect** statement from the following w.r.t dicot leaf.
- (1) Mesophyll is made up of parenchyma.
 - (2) Mesophyll has two types of cells - the palisade parenchyma and the spongy parenchyma.

- (3) The abaxially placed palisade parenchyma is made up of elongated cells.
- (4) Spongy parenchyma cells are oval or round.

115. **Assertion (A):** The mango trees cannot grow in temperate countries like Canada and Germany.

Reason (R): Tuna fish are abundantly found beyond tropical latitudes in the ocean.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

116. Identify the **incorrect** statement(s) w.r.t symmetry in flowers.

- A. The flowers of gulmohur are zygomorphic.
- B. The flowers of *Cassia* are actinomorphic.
- C. The flowers of chilli are zygomorphic.
- D. The flowers of mustard are actinomorphic.

Choose the *most appropriate* answer from the options given below:

- (1) A only
- (2) B and C only
- (3) B and D only
- (4) A, C and D only

117. In all the following plants, leaves of the juvenile plants are different in shape from those in mature plants, except:

- (1) cotton.
- (2) buttercup.
- (3) larkspur.
- (4) coriander.

118. Given below are two statements:

Statement I: There is no increase in the amount of DNA during interphase of cell cycle.

Statement II: In the G_2 phase, proteins for preparation of mitosis are synthesised.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

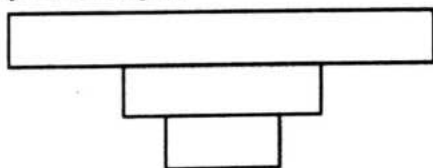
119. Which of the following components are essential for the process of transcription?

- (a) Ribosomes
- (b) RNA polymerase
- (c) DNA template
- (d) Amino acids
- (e) Promoter region

Choose the correct combination of components from the options given below:

- (1) (b), (c) and (e) only
- (2) (a), (b) and (d) only
- (3) (c), (d) and (e) only
- (4) (b), (d) and (e) only

120. The pyramid given below can be best illustrated by the example:



- (1) of biomass in sea.
- (2) of biomass in a grassland.
- (3) of energy in grassland.
- (4) of energy in sea.

121. Match List-I with List-II.

List-I		List-II	
(A)	Smooth endoplasmic reticulum	(I)	Involved in protein synthesis
(B)	Centriole	(II)	Made of tubulin protein fibres
(C)	Golgi complex	(III)	Dense reticular structure near the nucleus
(D)	Rough endoplasmic reticulum	(IV)	Involved in lipid synthesis

Choose the **correct** answer from the options given below:

- (1) A-IV, B-II, C-I, D-III
- (2) A-III, B-II, C-IV, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-IV, B-II, C-III, D-I

122. In biosynthetic phase of photosynthesis, the carbon dioxide acceptor in C_3 plants is:

- (1) phosphoenol pyruvate (PEP).
- (2) ribulose 1, 5-bisphosphate (RuBP).
- (3) phosphoglyceric acid (PGA).
- (4) ribulose 1, 3-bisphosphate (RuBP).

123. Choose the **incorrectly** matched pair.

- (1) Chasmogamous flowers - Anther and stigma are exposed
- (2) Cleistogamous flowers - Flower remains closed
- (3) Geitonogamy - Pollination between the flowers of same plant
- (4) Xenogamy - Pollination within the same flower

124. Identify the **correct** statement(s) from the following.

- A. Plant growth and development are under the control of both intrinsic and extrinsic factors.
- B. PGR is the only intrinsic factor that controls the plant growth and development.
- C. Cytokinin helps the plants to increase their absorption surface by promoting root growth and root hair formation.
- D. Ethylene helps in lateral shoot growth and adventitious shoot formation.

Choose the **most appropriate** answer from the options given below:

- (1) A only
- (2) A and C only
- (3) B and D only
- (4) A, B and D

125. Match List-I with List-II.

List-I	List-II
(A) Autosomal dominant trait	(I) Myotonic dystrophy
(B) Autosomal recessive trait	(II) Haemophilia
(C) Sex-linked recessive disorder	(III) Sickle-cell anaemia

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III
- (2) A-III, B-II, C-I
- (3) A-II, B-III, C-I
- (4) A-I, B-III, C-II

126. Identify the **incorrect** statement(s) from the following.

- A. More than 70 per cent of all the species recorded are animals.
- B. India now has 90 biosphere reserves.
- C. India now has 14 national parks and 48 wildlife sanctuaries.

D. Total number of biodiversity hotspots in the world is 34.

Choose the **most appropriate** answer from the options given below:

- (1) A only
- (2) B only
- (3) B and C
- (4) C and D

127. Identify the **incorrectly** matched pair.

- (1) Ribosome of chloroplast - 70S type
- (2) Ribosome of cytoplasm - 80S type
- (3) Plasmid DNA - Genomic DNA in prokaryotes
- (4) Plasmid DNA - Small circular DNA

128. Identify the reaction of glycolysis that is coupled with reduction of NAD^+ .

- (1) Glucose $\rightarrow$ Glucose 6-phosphate
- (2) Fructose 1, 6-diphosphate $\rightarrow$ PGAL + Dihydroxy acetone phosphate
- (3) Glucose 6-phosphate $\rightarrow$ Fructose 6-phosphate
- (4) 3-phosphoglyceraldehyde $\rightarrow$ 1, 3-bisphosphoglycerate

129. Following are the events that occur during meiosis:-

- (a) Compaction of chromosomes start
- (b) DNA replication
- (c) Meiotic spindle assembled
- (d) Dissolution of synaptonemal complex
- (e) Action of recombinase enzyme
- (f) Separation of sister chromatids
- (g) Segregation of homologous chromosomes
- (h) Formation of tetrad of cells

Which of the following is the correct sequence of events?

- (1) (b) $\rightarrow$ (a) $\rightarrow$ (e) $\rightarrow$ (d) $\rightarrow$ (c) $\rightarrow$ (f) $\rightarrow$ (g) $\rightarrow$ (h)
- (2) (a) $\rightarrow$ (b) $\rightarrow$ (e) $\rightarrow$ (d) $\rightarrow$ (c) $\rightarrow$ (f) $\rightarrow$ (g) $\rightarrow$ (h)
- (3) (b) $\rightarrow$ (a) $\rightarrow$ (c) $\rightarrow$ (d) $\rightarrow$ (f) $\rightarrow$ (e) $\rightarrow$ (g) $\rightarrow$ (h)
- (4) (b) $\rightarrow$ (a) $\rightarrow$ (e) $\rightarrow$ (d) $\rightarrow$ (c) $\rightarrow$ (g) $\rightarrow$ (f) $\rightarrow$ (h)

130. Identify the correct groups of components that describe the pollination mechanisms of the following aquatic plants:

(a) *Vallisneria*
 (b) *Hydrilla*
 (c) Water hyacinth
 (d) Water lily
 (e) *Zostera*

Choose the correct option from the following:

- (1) (a), (b), and (e) are water-pollinated; (c) and (d) are pollinated by insects or wind
 (2) (a), (c), and (e) are water-pollinated; (b) and (d) are pollinated by insects or wind
 (3) (b), (d), and (e) are water-pollinated; (a) and (c) are pollinated by insects or wind
 (4) (c), (d), and (e) are water-pollinated; (a) and (b) are pollinated by insects or wind

131. Given below are two statements:

Statement I: Equational division in somatic cells ensures conservation of the chromosome number of the parent in the daughter cell.

Statement II: Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

132. The family of mango is:

(1) Anacardiaceae
 (2) Malvaceae
 (3) Poaceae
 (4) Cruciferae

133. The minimum weight of tomatoes obtained from a plant is 40 g. The maximum weight of tomatoes obtained is 140 g. What will be the weight of tomatoes obtained by crossing plants with genotype AABB × aabb?

(1) 50 g (2) 70 g
 (3) 90 g (4) 25 g

134. Choose the **incorrect** statement from the following w.r.t bryophytes.

(1) The sex organs are multicellular.
 (2) The male sex organ is called antheridium.
 (3) They produce non-flagellated antherozoids.
 (4) The female sex organ is called archegonium.

135. Choose the **incorrect** statement from the following.

(1) When a cell loses the capacity to divide, it leads to differentiation.
 (2) Differentiation results in development of structures that are commensurate with the function the cells finally have to perform.
 (3) General principles for differentiation for cells, tissues and organs are different and unique.
 (4) A differentiated cell may dedifferentiate and then redifferentiate.

Botany : Section-B (Q. No. 136 to 150)

136. Identify the number of **correct** statements from the following.

A. Glucose is the favoured substrate for respiration.
 B. All fats are usually first converted into glucose before they are used for respiration.
 C. If fatty acids were to respire they would first be degraded to acetyl CoA.
 D. Glycerol would enter the respiratory pathway after being converted to PGA.

Choose the *most appropriate* answer from the options given below:

(1) 1 (2) 2
 (3) 3 (4) 4



137. $AaBbCcDD \times aabbccDD$

For the above cross, select the incorrect option.

- (1) The total number of male and female gametes formed is 9
- (2) It shows 1 : 1 : 1 : 1 genotypic ratio
- (3) The total number of phenotypes and genotypes formed is 8
- (4) It is a tetrahybrid cross

138. Histones are rich in _____ amino acid residues and _____.

- (1) basic; lysine and arginine
- (2) acidic; lysine and arginine
- (3) positive; lysine and glutamic acid
- (4) negative; lysine and glutamic acid

139. Identify the **correct** sequence of steps of Hershey-Chase experiment.

- (1) Infection → Blending → Centrifugation
- (2) Infection → Centrifugation → Blending
- (3) Blending → Infection → Centrifugation
- (4) Centrifugation → Blending → Infection

140. Identify the **incorrect** statement w.r.t meiosis.

- (1) Interkinesis is followed by prophase II.
- (2) G_2 phase is followed by prophase I.
- (3) During metaphase II, microtubules from opposite poles of the spindle get attached to the kinetochores of sister chromatids.
- (4) Anaphase I begins with the simultaneous splitting of the centromere of each chromosome.

141. Which of the following are key components used in the Calvin cycle's carboxylation phase?

- (a) Ribulose-1,5-bisphosphate
- (b) 3-phosphoglycerate
- (c) ATP
- (d) NADPH
- (e) CO_2

Choose the correct answer from the options given below:

- (1) (a), (b) and (c) only
- (2) (a), (d) and (e) only
- (3) (a), (e) only
- (4) (b), (c) and (d) only

142. Choose the **correct** example of ecological diversity.

- (1) *Rauwolfia vomitoria* growing in different Himalayan ranges with different potency and concentration of the active chemical (reserpine).
- (2) Western Ghats have a greater amphibian species diversity than the Eastern Ghats.
- (3) India has a greater ecosystem diversity than a Scandinavian country like Norway.
- (4) 50,000 genetically different strains of rice in India.

143. In bean flowers, the two smallest petals are known as:

- (1) standard. (2) wings.
- (3) lateral petals. (4) keel.

144. Identify the following as **true (T)** or **false (F)** statements w.r.t prophase I of meiosis.

- A. Bivalent or tetrad is more clearly visible in pachytene phase.
- B. Recombination nodules appears in the diplotene phase.
- C. Crossing over is an enzyme-mediated process and the enzyme involved is called recombinase.
- D. X-shaped structures formed during pachytene phase are called chiasmata.

A B C D

- (1) T T T T
- (2) T T F F
- (3) T F T F
- (4) F T F T

145. From the following, identify the number of structures that are present **only** in the prokaryotic cells.

Centriole, chloroplast, chromoplast, golgi bodies, plasmid, mesosome, ribosome, cell wall, nucleus

- (1) 6 (2) 5
- (3) 2 (4) 4

146. Identify the **incorrect** statement.

- (1) In bacteria, there are three major types of RNAs: mRNA (messenger RNA), tRNA (transfer RNA), and rRNA (ribosomal RNA).
- (2) The mRNA provides the template, tRNA brings amino acids and reads the genetic code, and rRNAs play structural and catalytic role during translation.
- (3) There are multiple DNA-dependent RNA polymerases that catalyse transcription of different types of RNA in bacteria.
- (4) RNA polymerase binds to the promoter and initiates transcription.

147. Match List-I with List-II.

List-I	List-II
(A) Adenine	(I) Purine
(B) Thymine	(II) Pyrimidine required in RNA synthesis
(C) Uracil	(III) Pyrimidine required in DNA synthesis
(D) Ribose	(IV) Sugar required in RNA synthesis

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
- (2) A-II, B-I, C-III, D-IV
- (3) A-I, B-III, C-II, D-IV
- (4) A-III, B-II, C-I, D-IV

148. Arrange the following locations based on the latitudinal gradients in order of increasing bird species diversity, starting from the least diverse:

- (a) Greenland (b) New York
- (c) Colombia (d) India

Choose the correct sequence from the options given below:

- (1) (a), (b), (d), (c)
- (2) (a), (b), (c), (d)
- (3) (d), (c), (b), (a)
- (4) (c), (d), (b), (a)

149. In snapdragon plants, broad leaf is completely dominant to narrow leaf whereas red flower colour is incompletely dominant to ivory (The genes for leaf width and flower colour are not linked). If plant which is heterozygous for both genes is crossed with a true-breeding broad-leaved red-flowered plant, then the expected proportion of broad-leaved plants with pink flowers amongst the offspring would be

- (1) 1 in 4 (2) 2 in 4
- (3) 3 in 4 (4) 4 in 4

150. Identify the **incorrect** match.

- (1) Ethylene - Initiates flowering in pineapples
- (2) Kinetin - Occurs naturally in plants
- (3) Gibberellic acids - Hasten maturity period of the plants
- (4) Auxins - Prevent fruit and leaf from dropping at early stage

Zoology : Section-A (Q. No. 151 to 185)

151. During proliferative phase:

- (1) myometrium wall is sloughed off.
- (2) endometrium wall is sloughed off.
- (3) estrogen stimulate proliferation of endometrium of uterus.
- (4) ovulation occurs.

152. Match List-I with List-II:

List-I	List-II
(A) Gout	(I) Decreased levels of estrogen.
(B) Osteoporosis	(II) Low Ca^{++} ions in the blood.
(C) Tetany	(III) Inflammatory disorder.
(D) Muscular dystrophy	(IV) Genetic disorder.

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
- (2) A-IV, B-III, C-I, D-II
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-I, C-III, D-IV



153. At which level of thoracic vertebra, the trachea divides into left and right primary bronchi?
- 2nd thoracic vertebra
 - 5th thoracic vertebra
 - 4th thoracic vertebra
 - 3rd thoracic vertebra
154. Arachidonic acid is a fatty acid which exhibits:
- 24 carbon atoms including carboxyl carbon.
 - 20 carbon atoms including carboxyl carbon.
 - 14 carbon atoms including carboxyl carbon.
 - 26 carbon atoms including carboxyl carbon.
155. Choose the **correctly** matched pair.
- Cartilage - Non pliable, hardest tissue
 - Ligament - Connect bone to bone
 - Areolar tissue - Dense regular connective tissue
 - Tendon - Loose connective tissue
156. Ultrafiltration 'occurs in which part of the nephron?
- Bowman's capsule
 - Loop of Henle
 - Proximal convoluted tubule
 - Distal convoluted tubule
157. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
- Assertion (A):** Memory-based acquired immunity evolved in higher vertebrates is based on the ability to differentiate foreign organisms (e.g., pathogens) from selfcells.
- Reason (R):** Rheumatoid arthritis is an auto-immune disease.
- In the light of the above statements, choose the **correct** answer from the options given below:
- A is true but R is false.
 - A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is NOT the correct explanation of A.
158. Inspiration occurs when intra-pulmonary pressure is:
- higher than atmospheric pressure.
 - lower than atmospheric pressure.
 - equal to atmospheric pressure.
 - zero compared to atmospheric pressure.
159. Select the **correct** statements w.r.t. frog.
- A. Frog's feet have webbed digits that help in swimming.
- B. The protective coloration in frogs is known as mimicry.
- C. Frogs exhibit sexual dimorphism.
- A and B only
 - A, B and C
 - B and C only
 - A and C only
160. What does RCH stand for?
- Reproductive and Complete Health programme
 - Regional and Central Health care programme.
 - Reproductive and Child Health care programme
 - Reproductive and Cumulative Health programme
161. The maximum amount of electrolytes and water (70 - 80%) from the glomerular filtrate is reabsorbed in which part of the nephron?
- Ascending limb of loop of Henle
 - Distal convoluted tubule
 - Proximal convoluted tubule
 - Descending limb of loop of Henle
162. Match List-I with List-II:
- | List-I | List-II |
|---------------------|--------------------------|
| (A) Flying squirrel | (I) Spotted cuscus |
| (B) Lemur | (II) Tasmanian tiger cat |
| (C) Bobcat | (III) Tasmanian wolf |
| (D) Wolf | (IV) Flying phalanger |

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-II, B-I, C-III, D-IV
- (3) A-IV, B-I, C-II, D-III
- (4) A-III, B-I, C-II, D-IV

163. Which of the following statements are **correct** for collagen?

- A. It is the most abundant protein in the whole biosphere.
- B. It functions as the intercellular ground substance.
- C. It is the most abundant protein in the animal world.
- D. It functions as the intracellular ground substance.

- (1) A and B only (2) B and C only
- (3) C and D only (4) A and D only

164. Which of the following is a natural method of contraception?

- (1) LNG-20
- (2) Condom
- (3) Periodic abstinence
- (4) Cu-T

165. Given below are two statements:

Statement I: In frogs, fertilisation is internal and takes place in water.

Statement II: The cloaca is a small, median chamber that is used to pass faecal matter, urine and sperms to the exterior.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

166. Read the following statements and choose the **correct** option.

- A. Infections which are transmitted through sexual intercourse are collectively called sexually transmitted infections.
- B. The chances of conception are almost nil as long as mother breast feeds her child fully.
- C. India was amongst the second countries in the world to initiate action plans and programmes at a national level.

- (1) A only
- (2) A and B only
- (3) B and C only
- (4) A, B and C

167. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: In cartilaginous joints, the bones involved are joined together with the help of cartilages.

Reason R: The joint between the adjacent vertebrae in the vertebral column is an example of cartilaginous joint and it permits no movements.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

168. Select the **odd** one w.r.t. early molecular diagnosis of diseases.

- (1) Recombinant DNA technology
- (2) ELISA
- (3) PCR
- (4) Serum analysis

169. Choose the **correct** statement from the following.
- (1) Flatworms are mostly endoparasites.
 - (2) In all members of phylum Arthropoda, excretion takes place through Malpighian tubule.
 - (3) All the general characters of *Nereis* are similar to earthworm due to this these both are placed into same phyla.
 - (4) Members of Mollusca have a segmented body which have distinct head, muscular foot and visceral hump.

170. Which of the following statements is **correct** about "crystal" proteins?

- (1) They are produced by the plant itself.
- (2) They are toxic to insects.
- (3) They are produced by viruses.
- (4) None of these

171. Match List-I with List-II.

List-I	List-II
(A) Male <i>Drosophila</i>	(I) XY
(B) Male Birds	(II) ZZ
(C) Female <i>Drosophila</i>	(III) XX
(D) Female Birds	(IV) ZW

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-I, D-II
- (4) A-I, B-II, C-IV, D-III

172. Read the following statements (A-D).

- A. The brain capacities of *Homo habilis* were between 650-800cc.
- B. The amphibians evolved into reptiles.
- C. *Ichthyosaurs* was fish like reptile.
- D. The original finches were seed-eating.

Which of the above statements are **correct**?

- (1) A, B and C only
- (2) A and B only
- (3) A, B and D only
- (4) A, B, C and D

173. Match List-I with List-II:

List-I	List-II
(A) Floating ribs	(I) Located between second and seventh ribs.
(B) Acromion	(II) Head of the humerus.
(C) Scapula	(III) Clavicle
(D) Glenoid cavity	(IV) Do not connect with the sternum.

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-III, B-II, C-IV, D-I
- (3) A-IV, B-III, C-I, D-II
- (4) A-II, B-IV, C-I, D-III

174. Which of the following is the **correct** sequence of Polymerase chain reaction (PCR)?

- (1) Denaturation → Annealing → Extension
- (2) Extension → Denaturation → Annealing
- (3) Annealing → Extension → Denaturation
- (4) Denaturation → Extension → Annealing

175. Complete the analogy.

Oxytocin : Parturition :: LH: _____.

Choose the **correct** option to fill the blank.

- (1) Ovulation
- (2) Lactation
- (3) Oogenesis
- (4) Implantation

176. Which of the following is responsible for transmitting impulses from one neuron to another?

- (1) Synaptic cleft
- (2) Neurotransmitter
- (3) Myelin sheath
- (4) Axon hillock

177. MALT constitutes about _____ percent of the lymphoid tissue in human body.

- (1) 50
- (2) 20
- (3) 70
- (4) 10

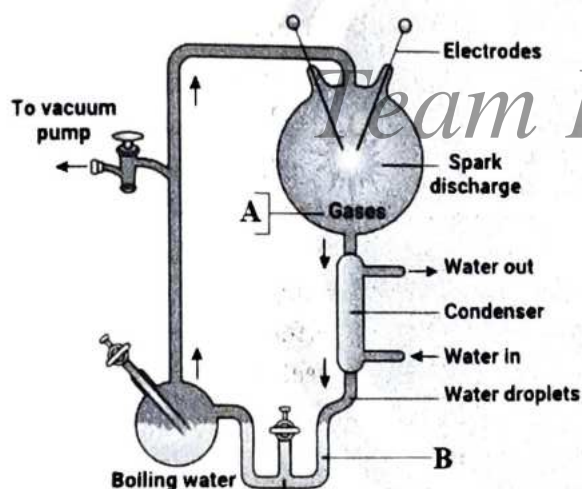
178. Match List-I with List-II:

List-I	List-II
(A) Diastole	(I) Contracted state.
(B) P-wave	(II) Depolarisation of the ventricles.
(C) Systole	(III) Relaxed state.
(D) QRS complex	(IV) Depolarisation state of atria.

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-III, B-IV, C-I, D-II
- (3) A-IV, B-III, C-I, D-II
- (4) A-I, B-II, C-III, D-IV

179. Observe the below given diagram of stanley Miller's experimental setup and identify A and B.



A	B
(1) CH_4 , NH_3 , H_2O , H_2 , O_2	Water containing inorganic compounds only.
(2) CH_4 , NH_3 , H_2O , H_2	Water containing organic compounds.
(3) CH_4 , H_2O	Water containing inorganic compounds.
(4) CH_4 , NH_3 , H_2O , H_2 , O_2	Water containing organic compounds.

180. The respiratory centre which regulates respiration is located in:

- (1) cerebellum.
- (2) medulla oblongata.
- (3) cerebrum.
- (4) hypothalamus.

181. Insulin produced by genetic engineering is used to treat:

- (1) cancer
- (2) diabetes
- (3) hypertension
- (4) asthma

182. Match List-I with List-II:

List-I	List-II
(A) Mons pubis	(I) Cavity of the cervix.
(B) Labia majora	(II) Paired folds of tissue in the form of lips under the labia majora.
(C) Labia minora	(III) Paired fleshy folds of tissue which extend down from mons pubis.
(D) Cervical canal	(IV) Fatty tissue covered by skin and pubic hairs.

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-II, B-IV, C-I, D-III
- (4) A-III, B-IV, C-I, D-II

183. Match List-I with List-II:

List-I	List-II
(A) Gills	(I) Pulmonary respiration
(B) Network of tubes	(II) Cutaneous respiration
(C) Lungs	(III) Tracheal respiration
(D) Moist skin	(IV) Branchial respiration

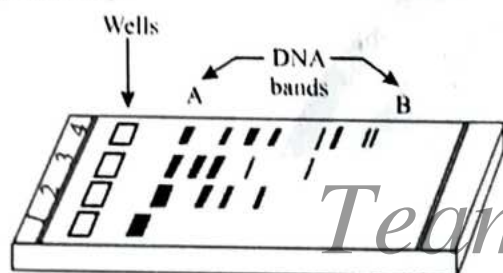
Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-IV, B-I, C-III, D-II
- (3) A-II, B-IV, C-I, D-III
- (4) A-IV, B-III, C-I, D-II

184. The immunity acquired after recovering from a disease is called:

- (1) active immunity (2) passive immunity
- (3) innate immunity (4) autoimmunity

185. Select the technique and type of DNA fragments (A and B).



	Technique	A	B
(1)	Recombinant DNA technology	Largest	Smallest
(2)	Recombinant DNA technology	Smallest	Largest
(3)	Agarose Gel Electrophoresis	Largest	Smallest
(4)	Agarose Gel Electrophoresis	Smallest	Largest

Zoology : Section-B (Q. No. 186 to 200)

186. Early symptoms of most STDs are:

- A. itching. B. fluid discharge.
- C. slight pain. D. swelling.

- (1) A, B and D only
- (2) A, B and C only
- (3) A, C and D only
- (4) A, B, C and D

187. Read the following statements.

- A. In 1997, an American company got patent rights on Basmati rice through the International Patent and Trademark Office.
- B. There are an estimated 50,000 varieties of rice in India alone.
- C. First clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase deficiency.
- D. In 1983, Eli Lilly company prepared DNA sequences corresponding to A, B and C chains for human insulin.

In the above statements how many are **incorrect** statements?

- (1) 4 (2) 3
- (3) 2 (4) None of these

188. Match List-I with List-II:

List-I (Component)	List-II (% of the total cellular mass)
(A) Water	(I) 3
(B) Proteins	(II) 70-90
(C) Carbohydrates	(III) 2
(D) Lipids	(IV) 10-15

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
- (2) A-II, B-IV, C-III, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-III, B-I, C-II, D-IV

189. Given below are two statements:

Statement I: Change in frequency of alleles in a population results in evolution.

Statement II: Evolution for Darwin was a gradual process.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

190. What is the primary function of the nictitating membrane in frogs?

- (1) Protects the eyes.
- (2) Assists in hearing.
- (3) Facilitates locomotion.
- (4) Aids in digestion.

191. Given below are two statements:

Statement I: MTPs are considered relatively safe during the second trimester.

Statement II: Infertility cases due to inability of the male partner to inseminate the female could be corrected by artificial insemination technique. In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

192. The cerebrum is made up of _____ cerebral hemisphere.

- (1) 1 (2) 2
- (3) 3 (4) 4

193. Which of the following statements are **correct**?

- A. In earthworm, the body shows metameric segmentation.
- B. Aschelminthes are pseudocoelomates.
- C. Sponges are mostly asymmetrical.
- D. Mesoglea is an undifferentiated layer present in between the ectoderm and the endoderm.

- (1) A and B only
- (2) B and C only
- (3) B, C and D only
- (4) All of these

194. A person has protruding eyes, weight loss and higher body temperature, he is suffering from:

- (1) cretinism.
- (2) diabetes.
- (3) Grave's disease.
- (4) acromegaly.

195. What is the primary function of the seminal vesicles in the male reproductive system?

- (1) To produce sperm.
- (2) To secrete a fluid that nourishes and helps transport sperm.
- (3) To produce testosterone.
- (4) To regulate the temperature of the testes.

196. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R: **Assertion A:** Left ventricle possesses the thickest wall among all heart chambers.

Reason R: Left atrium receives the deoxygenated blood from lungs.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

197. Hormones stored and released from neurohypophysis are:

- (1) oxytocin and vasopressin.
- (2) follicle stimulating hormone and leutinizing hormone.
- (3) prolactin and vasopressin.
- (4) thyroid stimulating hormone and oxytocin.

198. Match List-I with List-II:

List-I		List-II	
(A)	ELISA	(I)	Conventional method
(B)	PCR	(II)	Antigen antibody interaction
(C)	r-DNA technology	(III)	Golden rice
(D)	Urine analysis	(IV)	Early detection of HIV.

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-II, B-IV, C-III, D-I
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-IV, C-I, D-II

199. Vaccination provides immunity by:

- (1) producing antibodies directly in the bloodstream.
- (2) introducing a weakened or killed pathogen to stimulate the immune system.
- (3) increasing the number of red blood cells.
- (4) blocking the entry of pathogens.

200. What is the main excretory product in birds and reptiles that conserves water?

- (1) Urea
- (2) Ammonia
- (3) Uric acid
- (4) None of these

Team Kohinoor